# Detroit Council Districts 1 - 7 Regression Model Comparison: R | SPSS 2023 - 2025 Sales

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## Contents



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### Introduction

This report contains multiple comparisons of linear regression models prepared in SPSS, and R. In order to assure model reproducibility, datasets and methods utilized in SPSS were implemented in R as closely as possible. Due to the nature of stepwise backward elimination however, exact thresholds for predictor removal could not be fine tuned for every model designed for all 7 districts. Overall, models created in R using the same parameters as those in SPSS had exceptionally close performance metrics.

These results demonstrate that the current linear regression model approach is viable across platforms.

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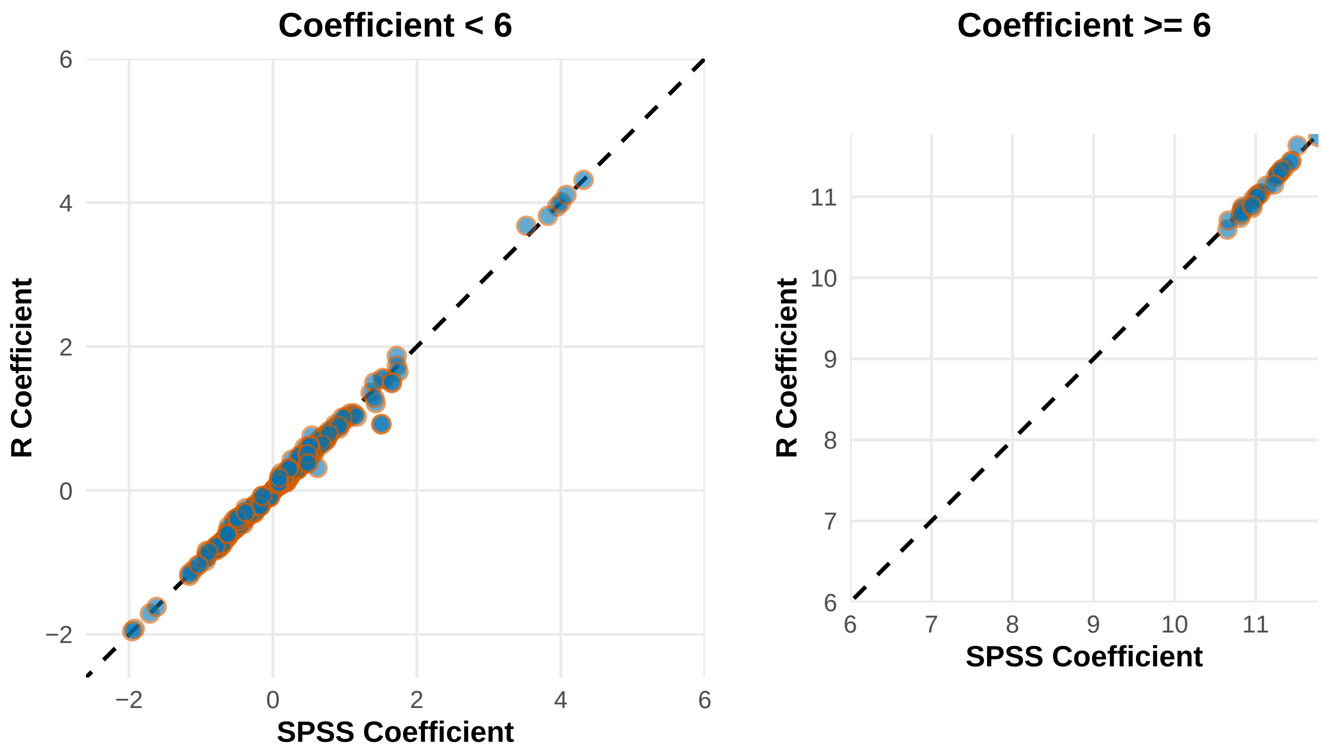
### Coefficient-Concordance

The coefficients represented on the graph below indicate: 1) The direction, and 2) The impact of a predictor on the dependent variable (for these models home value). Values close to zero have a negligible impact on sale price. Strong negative values correlate with decreased sale price (e.g., unsound condition properties), high positive values correlated with increased sale price (e.g., properties in good condition).

As SPSS and R coefficients overlap significantly, they produce a line with slope = 1 when plotted on opposite axes.

## SPSS | R Coefficient Concordance

Points on the dashed 1:1 line indicate equivalent coefficient estimates

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### Heatmap Overview

The following figure highlights the similarity (or difference) of a metric from an R model compared to the SPSS created equivalent.

## R | SPSS Performance Metric Comparison

Blue = R higher; orange = R lower; cell text shows R − SPSS

| | $R^2$ | Adj. $R^2$ | SEE | F | Median Ratio | COD | PRD | PRB | Within 10% | Within 20% | Within 50% |
|---|---|---|---|---|---|---|---|---|---|---|---|
| District 1 – Model 1 | −0.0002 | −0.0002 | +0.0000 | +0.0002 | NA | NA | NA | NA | NA | NA | NA |
| District 1 – Model 2 | +0.0003 | −0.0000 | −0.0000 | −0.0002 | NA | NA | NA | NA | NA | NA | NA |
| District 1 – Model 3 | −0.0001 | −0.0003 | +0.0000 | −0.0005 | +0.0002 | −0.00 | −0.0001 | NA | +4.51 | +9.59 | +19.90 |
| District 1 – Model 4 | +0.0005 | −0.0002 | +0.0000 | −0.0003 | −0.0004 | +0.00 | −0.0003 | +0.0000 | +5.53 | +10.89 | +23.69 |
| District 1 – Model 5 | −0.0000 | +0.0005 | −0.0000 | +0.0004 | −0.0002 | +0.00 | −0.0003 | −0.0005 | +5.48 | +10.79 | +23.74 |
| District 2 – Model 1 | +0.0000 | +0.0001 | −0.0000 | +0.0002 | NA | NA | NA | NA | NA | NA | NA |
| District 2 – Model 2 | −0.0003 | +0.0004 | −0.0000 | −0.0004 | NA | NA | NA | NA | NA | NA | NA |
| District 2 – Model 3 | +0.0000 | −0.0001 | −0.0000 | −0.0002 | −0.0490 | −0.01 | −0.0225 | NA | −0.49 | +0.66 | +1.48 |
| District 2 – Model 4 | +0.0002 | +0.0000 | −0.0000 | −0.0004 | −0.0371 | +0.00 | −0.0106 | +0.0350 | −0.09 | +1.12 | +0.54 |
| District 2 – Model 5 | −0.0196 | −0.0199 | +0.0155 | −6.3779 | −0.0284 | +0.01 | +0.0028 | +0.0223 | −1.28 | +0.05 | −0.36 |
| District 3 – Model 1 | −0.0011 | −0.0008 | +0.0003 | −0.1187 | NA | NA | NA | NA | NA | NA | NA |
| District 3 – Model 2 | +0.0000 | +0.0000 | +0.0000 | −0.0114 | NA | NA | NA | NA | NA | NA | NA |
| District 3 – Model 3 | −0.0001 | −0.0007 | +0.0001 | −0.0422 | −0.0006 | −0.00 | +0.0004 | NA | +4.34 | +7.43 | +20.10 |
| District 3 – Model 4 | +0.0005 | −0.0001 | −0.0030 | −1.2069 | +0.0030 | −0.00 | −0.0025 | +0.0054 | +4.77 | +10.18 | +24.28 |
| District 3 – Model 5 | +0.0003 | +0.0005 | −0.0022 | −1.2911 | −0.0009 | −0.00 | −0.0018 | +0.0045 | +3.81 | +11.70 | +24.12 |
| District 4 – Model 1 | −0.0001 | +0.0002 | −0.0000 | −0.0000 | NA | NA | NA | NA | NA | NA | NA |
| District 4 – Model 2 | −0.0004 | +0.0004 | +0.0000 | −0.0004 | NA | NA | NA | NA | NA | NA | NA |
| District 4 – Model 3 | +0.0004 | +0.0005 | −0.0000 | +0.0084 | −0.0248 | −0.01 | −0.0162 | NA | +0.92 | +1.31 | +1.84 |
| District 4 – Model 4 | +0.0004 | −0.0003 | −0.0000 | +0.0092 | −0.0166 | −0.00 | −0.0060 | +0.0038 | +1.40 | +0.66 | +0.33 |
| District 4 – Model 5 | +0.0000 | −0.0003 | −0.0000 | −0.0087 | −0.0156 | −0.00 | −0.0053 | +0.0035 | +1.47 | +0.79 | +0.33 |
| District 5 – Model 1 | −0.0002 | +0.0000 | −0.0000 | −0.0004 | NA | NA | NA | NA | NA | NA | NA |
| District 5 – Model 2 | −0.0004 | −0.0003 | +0.0000 | −0.0005 | NA | NA | NA | NA | NA | NA | NA |
| District 5 – Model 3 | −0.0000 | +0.0004 | +0.0000 | −0.0003 | −0.0197 | −0.13 | −0.1094 | NA | +4.31 | +3.95 | +6.54 |
| District 5 – Model 4 | −0.0023 | −0.0020 | −0.0014 | +0.7948 | +0.0000 | −0.07 | −0.0483 | +0.0184 | +2.81 | +4.22 | +4.53 |
| District 5 – Model 5 | −0.0069 | −0.0054 | +0.0015 | +1.9642 | +0.0034 | −0.07 | −0.0396 | +0.0128 | +5.09 | +5.00 | +4.71 |
| District 6 – Model 1 | +0.0024 | −0.0020 | +0.0005 | −0.3159 | NA | NA | NA | NA | NA | NA | NA |
| District 6 – Model 2 | −0.0135 | −0.0172 | −0.0024 | −2.1126 | NA | NA | NA | NA | NA | NA | NA |
| District 6 – Model 3 | −0.0100 | −0.0139 | −0.0022 | −1.8525 | −0.0360 | −0.06 | −0.0745 | NA | +2.18 | +3.94 | +4.11 |
| District 6 – Model 4 | +0.0073 | +0.0070 | +0.0036 | −0.2070 | −0.0255 | −0.04 | −0.0200 | +0.0006 | +1.35 | +1.76 | +6.09 |
| District 6 – Model 5 | +0.0052 | +0.0026 | +0.0057 | −1.5334 | −0.0091 | −0.05 | −0.0197 | −0.0046 | +3.78 | +7.63 | +4.07 |
| District 7 – Model 1 | −0.0039 | −0.0053 | +0.0014 | −2.7721 | NA | NA | NA | NA | NA | NA | NA |
| District 7 – Model 2 | −0.0091 | −0.0096 | +0.0027 | −1.7953 | NA | NA | NA | NA | NA | NA | NA |
| District 7 – Model 3 | −0.0052 | −0.0048 | +0.0015 | −0.7324 | −0.0035 | +0.00 | +0.0011 | NA | +5.85 | +11.89 | +23.71 |
| District 7 – Model 4 | −0.0324 | −0.0331 | +0.0080 | −6.9844 | −0.0109 | +0.01 | +0.0062 | −0.0439 | +6.58 | +13.36 | +27.61 |
| District 7 – Model 5 | −0.0344 | −0.0352 | +0.0084 | −7.0816 | −0.0075 | +0.01 | +0.0060 | −0.0495 | +6.95 | +13.09 | +27.31 |

R relative to SPSS

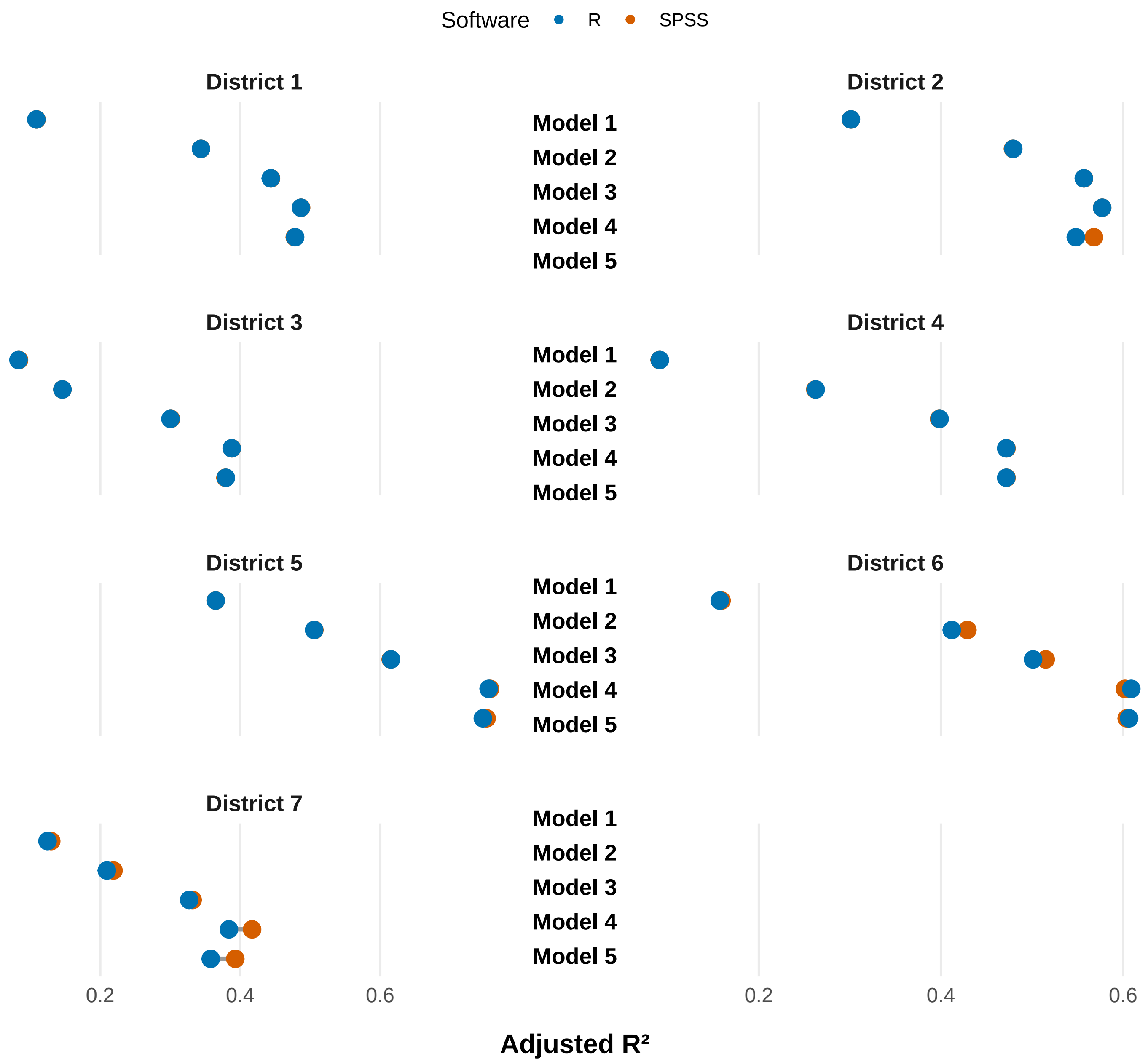
−1.0 −0.5 0.0 0.5 1.0

## Metric Comparison

### Adjusted R^2

## Adjusted $R^2$: SPSS | R

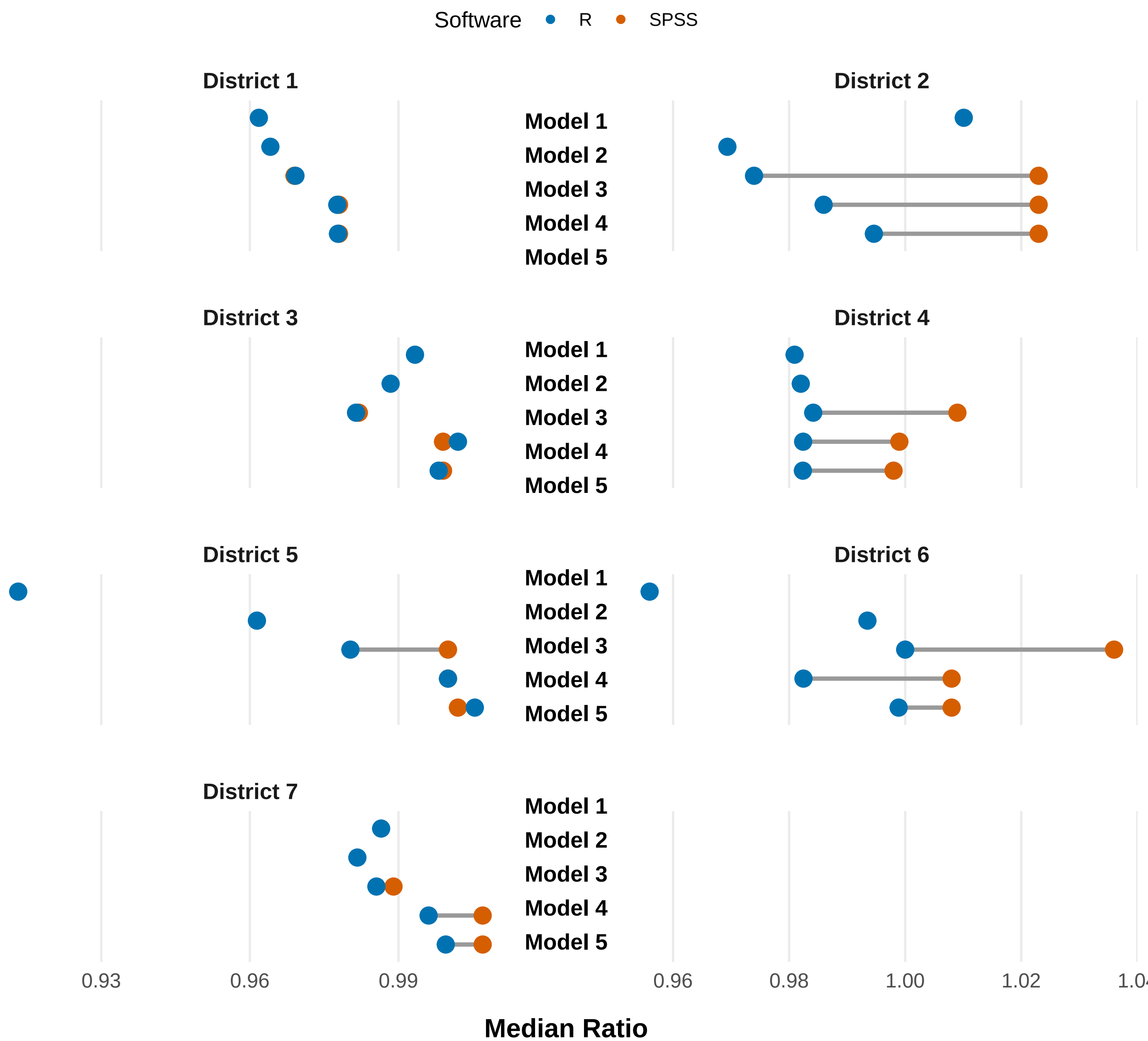
Each line connects the SPSS and R result for the same AVM



**Median Ratio**

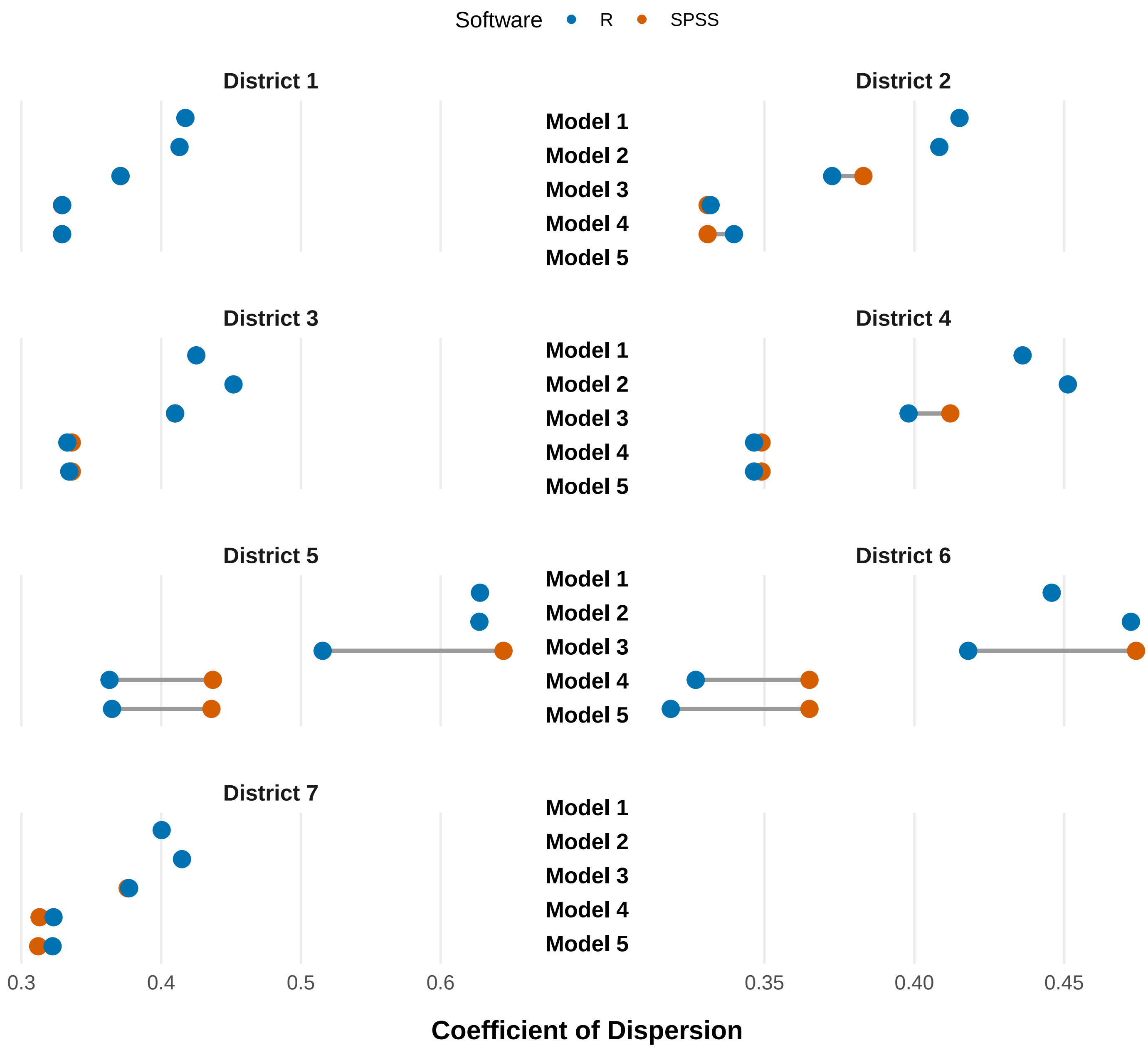
## Median Ratio: SPSS | R

Each line connects the SPSS and R result for the same AVM



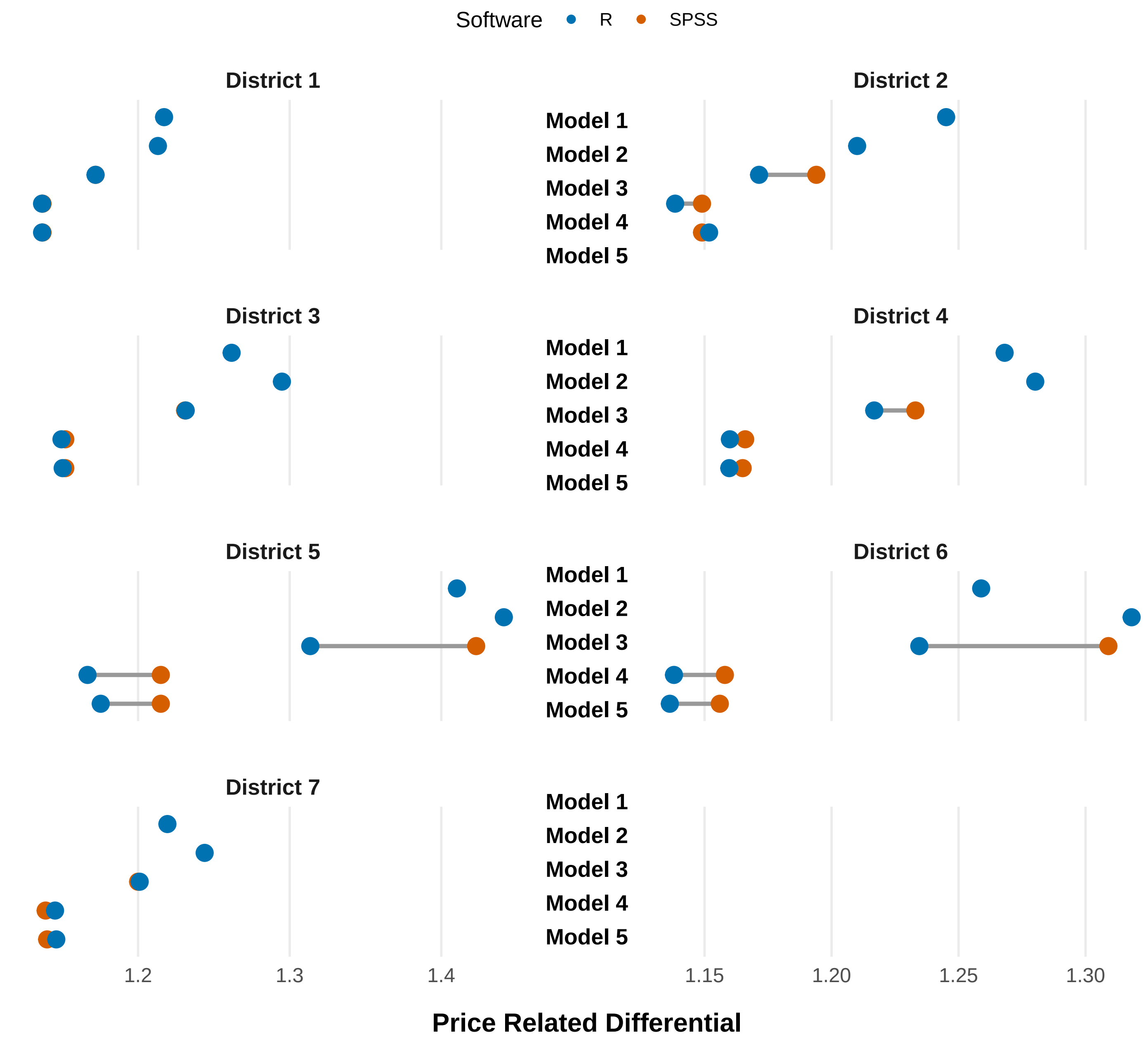
## Coefficient of Dispersion: SPSS | R

Each line connects the SPSS and R result for the same AVM



## Price Related Differential: SPSS | R

Each line connects the SPSS and R result for the same AVM



**Price Related Bias**

## Price Related Bias: SPSS | R

Each line connects the SPSS and R result for the same AVM